Beyond Horizontal Location Context: Measuring Elevation Using Smartphone's Barometer

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Abstract
Accurate estimation of elevation is important for many location based services. Although, it is possible to obtain altitude from GPS, its accuracy is unreliable and applicable in outdoors only. It is possible to use barometers on smartphones to estimate elevation in both indoor and outdoor scenarios. To this end, we proposed an integrated framework to provide ubiquitous and accurate elevation measurement using smartphones. Experiments conducted in both indoor and outdoor with different geographical characteristics reveal that our system can provide elevation with an error less than 5 meters in 90 % of the cases and less than 3 meters in 75 % of the cases, which is sufficient for most practical applications.

Author Keywords
Elevation measurement, Barometer, Smartphone sensors, Altitude, Floor localization

ACM Classification Keywords
H.4 [Information Systems Applications]: Miscellaneous

Introduction
Location context plays an indispensable role in context aware services. Until now researchers have mainly focused on localization on a 2D plane (latitude and

longitude). However, altitude or elevation is also important for location based services(LBS) including 3D navigation, indoor localization (e.g. floor), mountaineering, emergency rescue operations and so on. Although GPS can provide altitude,its accuracy is notoriously poor and unstable (sometimes the error can reach up to 2.5 times more than the horizontal one) [7]. Moreover, it does not work in indoor environments. Therefore, the objective of this research work is to develop a ubiquitous, less costly and highly accurate elevation measurement system.

Our method is based on a well established theory that air pressure (atmospheric pressure) change corresponds to elevation changes. Air pressure can be measured using a barometer. Based on this law, altimeters used in aircrafts use barometer to get the air pressure and use this information to calculate the altitude. They also need local air pressure information from control towers as reference points. These reference points can be used to calculate the elevation with acceptable accuracy only when one is within certain distance from that reference point.

Thanks to the development of sensor technologies, air pressure sensors are built-in in most smartphones nowadays. However, we have to tackle certain challenges before using these low cost embedded barometers to get accurate measurements of elevation. In a standalone mode (elevation measured using smartphone's barometer only,based on ISA (International Standard Atmosphere) model [14]), the error reaches tens of meters or even hundreds of meters (depending on weather conditions and altitude). Even other research works [10] using meteorological stations as reference points (as used by aircrafts), reported errors in the range of several meters to tens of meters. In this paper we propose a system called iBaro-altimeter that adresses this shortfall by solving 3 sub-problems.

The first problem we address is related to the density of reference points. Accuracy of elevation measurement at a point depends on the distance between this point and the reference point. However, the only reference points available are meteorological stations which are often sparsely located. Hence, we propose methods to introduce ad-hoc reference points. The second problem arises from the fact that these meteorological stations broadcast periodically, usually at one hour interval. However, within this period, the air pressure might change significantly. We integrate information from multiple reference points including the ad-hoc ones and also use a forecast model to estimate the air pressure on demand. The third problem we tackle is related to the accuracy of the barometer sensors embedded in smarphones. Earlier research work [8] showed that these sensors are less accurate and less stable than the barometers certified for professional use. We use multiple filtering techniques such as Kalman filter to handle this issue.

Our main contributions are as follows:

- We proposed an integrated framework to provide accurate elevation measurement which uses only smartphones without special infrastructure.
- We conducted a series of experiments to evaluate our system. Considering real application situations, we designed the following 3 scenarios for experiments: outdoor walking, mountain climbing and inside buildings.

System Overview

Theory of elevation measurement from air pressure
The relation between air pressure and elevation has been well studied, and it is defined as:

$$P_h = P_b \cdot \left[\frac{T_b}{T_b + L \cdot (Z_h - Z_b)}\right]^{\left[\frac{g \cdot M}{R^* \cdot L}\right]} \quad (1)$$

where

Z_h, Z_b: height above sea level (meter)
P_h: air pressure (pascal) at height Z_h
P_b: air pressure (pascal) at height Z_b
T_b: standard air temperature (K)
L: temperature lapse rate (K/m)
g: gravitational acceleration (m/s^2)
M: molar mass of earth's air $(kg/kmol)$
R^*: universal gas constant for air $(N * m/(kmol * K))$

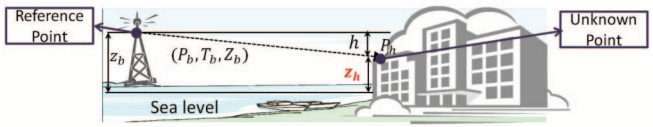


Figure 1: Elevation measurement from air pressure

To calculate elevation, we can inverse equation 1 as:

$$Z_h = Z_b + \frac{T_b}{L}\left[\left(\frac{P_h}{P_b}\right)^{-\frac{L \cdot R}{g}} - 1\right] \quad (2)$$

where $R = \frac{R^*}{M}$.

As shown in Figure 1, if status (P_b, T_b, Z_b) at a reference point is known and air pressure (P_h) is known too, it is easy to obtain the elevation (Z_h) from Equation 2 and relative height (by $h = Z_h - Z_b$).

Sensitivity analysis

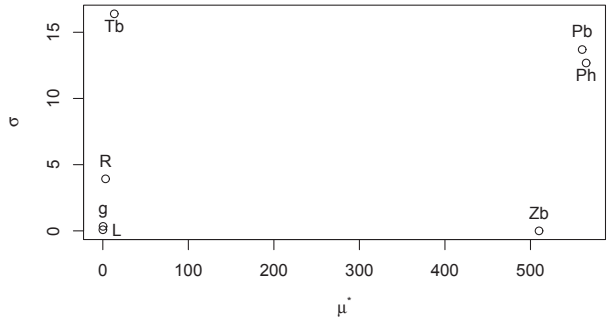


Figure 2: Sensitivity of input parameters of baro-altimeter equation

The challenge in elevation measurement is that there are always errors due to inaccurate measuremtns associated with each of the input parameters in Equation 2. Therefore we applied elementary effects method (of sensitivity analysis) to simulate the influence of parameters on the result. Elementary effects (EE) method [13] can identify non-influential inputs and rank them in order of importance. EE of Compolongo [2] provides two sensitivity measures for each input parameter:

- μ^*, assesses the overall importance of an input on the model output.
- σ, describes non-linear effects and interactions.

Low values of both $\mu*$ and σ correspond to a non-influent input. Figure 2 shows the result, which suggest that P_b and P_h are the most influential inputs, Z_b is also important, while g, L and R are the least influential inputs, T_b is less influential but affects the variability substantially. Consequently, in order to

enhance accuracy of elevation (Equation 2), we focus on reducing uncertainty of P_h, P_b, Z_b, T_b.

System architecture
Since there are meteorological stations for weather forecast service in most countries, it is a common practice to use them as reference points. However, these stations are sparsely located and they broadcast the information once every hour. This coarse spatial density (about tens of kilometers) and low update frequency (about once an hour) results in lower accuracy when elevation is obtained from barometer.

Here we propose a new system called *iBaro-altimeter* to resolve this problem. In our system we incorporate 3 types of reference points.

- *Base station*:
  These are meteorological stations usually provided by national meteorological agency.
- *DEM reference*:
  It is based on Digital Elevation Model (DEM). When a person (with smartphone) is on the ground in outdoor, it is possible to get the elevation from DEM map based on his current location (from GPS). When combined with air pressure measurement from the same smartphone, this status can be used as a temporal reference point.
- *Smartphone reference*:
  It is based on a smartphone within our system. If elevation of the smartphone is obtained from our system, then this smartphone itself can be used as an ad-hoc reference point. To avoid error accumulation, the *smartphone reference* cannot be referred to itself.

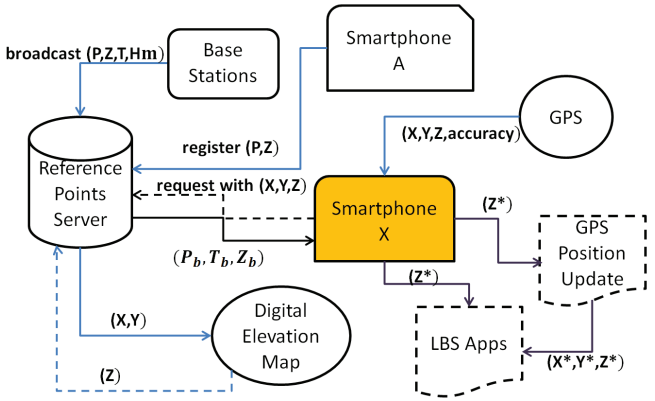


Figure 3: iBaro-altimeter architecture (X: longitude, Y: latitude, Z: altitude(elevation), P: air pressure, T: temperature)

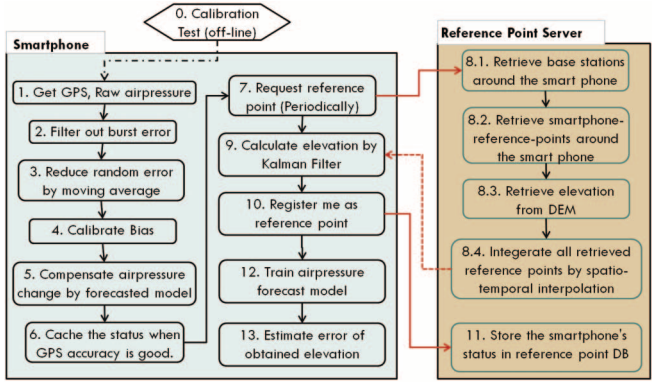


Figure 4: Workflow of iBaro-altimeter

Figure 3 shows the architecture of *iBaro-altimeter*. To calculate elevation of smartphone X, first it needs to get GPS information (latitude, longitude, altitude, accuracy). Then, using this information it requests the reference points server. On the other end, the server

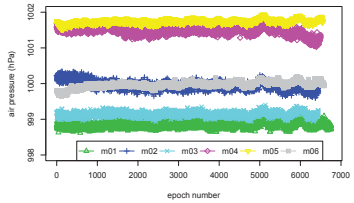


Figure 5: Difference in readings among pressure sensors

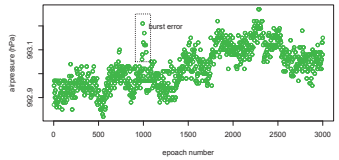


Figure 6: Random error and burst error of barometer

will integrate all reference points within a specified spatio-temporal radius around the smartphone and send the resulting reference point back to the smartphone. Finally, based on this virtual reference point, the smartphone can obtain accurate elevation which can be used in other LBS applications. Furthermore,it can also be used to improve GPS accuracy. Figure 4 shows how *iBaro-altimeter* works in a client/server fashion while details of major components are provided in the next section.

Major Components of the System

Sensor calibration

Barometers built in smartphones are not as accurate as the measurement devices of meteorological stations. According to the specification of smartphone's barometers, there is a maximum absolute error of 4 hPa. We do calibration test to eliminate this systematic error before using the smartphone's barometer. Figure 5 shows the pressure sensor readings in 6 smartphones at identical location and time. We found that the systematic error is different, but it is stable over longer period of time. Calibration test can be done when the elevation Z_h and Z_b, T_b, P_b are known exactly. By repeating the test, standard deviation can be obtained for evaluating the stability of barometer. The calibrated bias will be added before using the raw value of barometers.

Filtering burst error and random error

As can be seen in Figure 6 there are random errors and burst errors (spikes) in raw values of barometer (sampling rate: 2 Hz). According to the specifications of the barometer, there is a random error of less than ± 0.1 hPa (depends on models). It is easy to reduce this random error by a low pass filter or simply taking average.

On the other hand, burst error which is caused by electronic defect or drastic change in air pressure, can be identified by a threshold-based method. Threshold based method might not completely remove all burst error (depends on how we define burst error), nevertheless burst error will be further addressed at a later stage with adaptive Kalman filter.

Integrating reference points

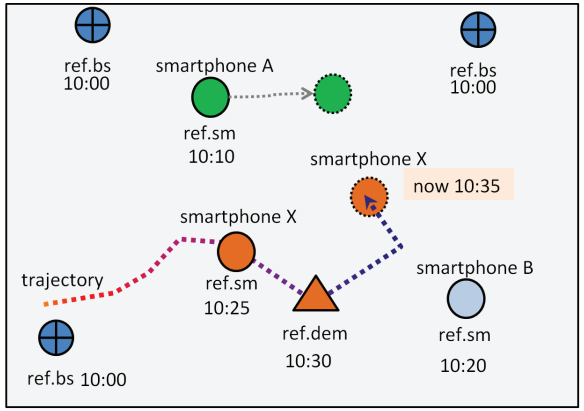


Figure 7: Reference points with dynamic spatio-temporal feature

In our system, we introduce two types of dynamic reference points(i.e. *DEM reference* and *smartphone reference*), and fixed *base stations*. As shown in Figure 7, *base stations* are stationary and broadcast information at fixed time interval, while *smartphone references* are mobile and update information dynamically, and *DEM reference* is available when GPS signal is very accurate. To estimate the elevation of $smartphoneX$, it is obviously not best to use the

nearest or the latest reference point, because all reference points are not fully accurate, meaning using a single point results in error accumulation. As such, we take into account all the reference points which are within a specified distance and time period (e.g. within 50 km and 1 hour until now). When a reference point is closer to the smartphone, the accuracy of elevation measurement is better, and the newer the reference information, the more accurate the measurement. In addition, the vertical distance between reference point and smartphone affects the accuracy too.

Each factor (horizontal distance, vertical distance and time gap) independently leads to error accumulation in a certain form (let's say $f(hs), g(vs), h(t)$ respectively). We assume their form follows an exponential function (Gaussian distribution) which can be verified by approximating the error distribution with regression method. Each reference point is given a weight w based on spatio-temporal distance as follows:

$$w = f(hs) * g(vs) * h(t) \tag{3}$$

where we assume $f(hs)$, $g(vs)$ and $h(t)$ follow $N(0, \sigma_{hs}^2)$, $N(0, \sigma_{vs}^2)$, $N(0, \sigma_t^2)$ respectively, then

$$w = (2\pi)^{-\frac{3}{2}}(\sigma_t \sigma_{hs} \sigma_{vs})^{-\frac{1}{2}} e^{-\frac{1}{2}[\frac{t^2}{\sigma_t^2} + \frac{{hs}^2}{\sigma_{hs}^2} + \frac{{vs}^2}{\sigma_{vs}^2}]} \tag{4}$$

where t is the time gap between the reference information time-stamp and current time, hs is the horizontal distance (calculated from latitude and longitude), vs is the vertical distance (altitude) between the reference point to the average elevation of base stations. We also assume that the effect of a factor should be marginal,when a factor reaches a specific value. Therefore, all these 3 factors will be scaled to [0,1] by dividing by a specific value.

In addition, although these 3 factors are considered independent, they are supposed to contribute to the weight at different rate of change. For example, one unit (say 1 minute) of time gap may not lead to same error as one unit (say 1 Km) of horizontal distance. This rate of change is related the variance (σ^2), therefore we can control it by adjusting σ_t^2, σ_{hs}^2 and σ_{vs}^2. From history of air pressure in different meteorological stations, we can estimate how much the air pressure changes with respect to specific time interval and distance.

In our system, heterogeneous reference points(static and dynamic) coexist. Different types of reference points have different accuracy due to their intrinsic characteristics. Therefore we need to assign different weight according to the type of reference points. Base stations usually perform good and are stable. We can give them a high constant credit ($credit.bs$). DEM reference has error due to the DEM map generation process, thus we give them a constant credit $credit.dem$ less than $credit.bs$. Moreover, DEM reference's credit also depends on the geographic feature where the target is located when retrieving it. Smartphone reference is not very reliable due to the cheap barometer's performance, thus we give a constant credit $credit.sm$ less than $credit.dem$. Moreover, Smartphone reference's credit also depends on the situation when its elevation is calculated. If the smartphone obtained the elevation from a lot of high reliable reference points, then the credit will be high.

Short-term forecast of air pressure

Since reference points are not updated frequently and requesting the reference points server is expensive (in terms of both communication and computation), the smartphone only requests the server at specific time

intervals. Moreover, it is reasonable to predict the air pressure in a short time period (less than 30 minutes). Consequently, we compensate the air pressure which is caused by weather change within the interval of reference request.

Because we only predict the air pressure over a short time around a certain location, it is reasonable to deal with air pressure as time series data and ignore spatial variations. Besides, we don't consider these seasonal patterns because their effect is negligible over short time periods. There are many forecasting models for time series data, including ARIMA, Holt-Winters method, exponential trend method and so on [6]. We applied additive dampened trend method because it is flexible (for details see [6]).

Estimating elevation
Although it is workable to directly calculate the elevation (Z_h) from reference point's information (T_b, P_b, Z_b) and smartphone's barometer (P_h), the elevations of trajectory (called elevation profile) are not stable because smartphone's barometer is too sensitive to the environment. As such, the obtained elevation looks jerky, which does not follow the natural movement of users (human). To tackle this problem, we apply an adaptive extended Kalman filter. Considering elevation measurement is a non-linear system, we don't know the true elevation, but can estimate it from the observation (air pressure). The states ($\mathbf{x}$) we are concerned with are elevation Z_h and elevation velocity $\dot{Z}_h$, the measurement ($\mathbf{z}$) is only air pressure (P_h). For process model, we assume it follows Newton's equation of motion. The measurement model is based on Equation 1.

As shown in Equation 2, the measurement (air pressure) relationship to the process (elevation) is non-linear, thus we have to apply Extended Kalman Filter (for details see [17]) in our system. In addition, we simplify the movement of human by Newton's equation of motion, although this model cannot fully match the movement (in elevation). As such we apply an adaptive Kalman filter as introduced in [12] to adjust the process noise.

Evaluation and Discussion

We conducted a series of experiments to evaluate our method. Experiments consisted of the following 3 scenarios: outdoor, mountain and indoor. Since we cannot acquire actual ground truth data, we compare our obtained elevation with DEM when the GPS accuracy is better than 6 meters. For base station reference points, we take advantage of meteorological stations of Japan Meteorological Agency. We use different brands of smartphones (Samsung S3, S4 and LG Nexus 4) for experiments. In most experiments, air pressure from barometer is sampled in 20 Hz, and elevation is estimated every second. The error (difference between DEM elevation and estimated elevation by our method) is measured by MAE (mean absolute error), RMSE (root mean squared error) and MRE (mean relative error).

Case 1: Outdoor walking
In this case, 5 experiment participants with 10 smartphones walked in Setagaya district, Tokyo. Eeach participant walked about 2 hours on different routes.

Figure 8 shows the MAE on different smartphones (different walking routes). **KF** is the elevation obtained by applying Kalman filter while **DR** is the elevation obtained without Kalman filter (other processes are

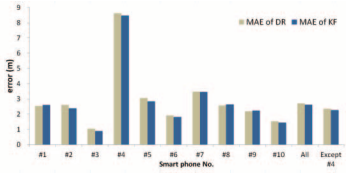


Figure 8: Elevation MAE on different smartphones by DR and KF

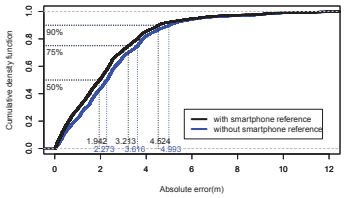


Figure 9: the CDF of elevation AE in Outdoor walking

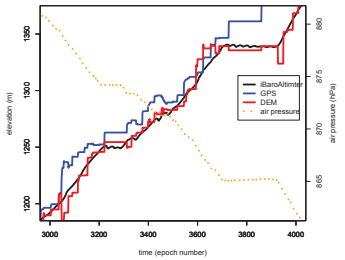


Figure 10: A part of elevation in mountain climbing

same). The label 'All' in the figure means the average of all smartphones, and the label 'Except #4' is the average of all smartphones except for smartphone #4. The MAE of most cases is very small (about 3 meters) but smartphone ♯4 is very large (about 9 meters). In fact, we found in the calibration test the variance of the barometer of smartphone #4 is very large, which is the reason why its result is abnormal. To avoid the error propagation of such malfunctioning barometer, the reference points server doesn't accept it as reference point. In addition, the result shows that **the elevation obtained by Kalman filter is generally better than without Kalman filter**. RMSE and MRE are not shown here, but they are less than 3.6 meters and 5% respectively.

Figure 9 shows the cumulative density of absolute error (AE) in all smart phones. This reveals that AE is less than 3.213 meters in 75% of the cases and less than 4.524 meters in 90%. Besides, we want to explore accuracy of our method if we don't use any smartphone reference point. Evaluation results show that AE without smartphone reference is less than 3.616 meters in 75% and less than 4.993 meters in 90%, **meaning integration of smartphone reference improved accuracy by about 13%**. However, it is worth noting that the contribution to accuracy by smartphone reference heavily depends on the setting of parameter $credit.sm$.

Case 2: Mountain climbing

In this scenario, experiment participants climbed 3 mountains (says $Mt\#1, Mt\#2, Mt\#3$ for convenience) on different days. For $Mt\#1$ and $Mt\#2$, we did not have any smartphone reference. Results in Table 1 show that MAE on each of the mountains is very small (less than 6 meters), and MRE is less than 1%. In addition, we compared the difference if DEM reference is integrated or not. In the case of $Mt\#1$ and $Mt\#2$ (table 1), **we found integration of DEM reference has advantage to a certain degree**. Similar to smartphone reference, the degree of accuracy improvement depends on setting of the parameter $credit.dem$. Note that, although the error of $Mt\#3$ (in which smartphone reference is used) is worse than other two mountains, it doesn't indicate that integrating smartphone reference works negatively, because the accuracy depends on the interaction of multiple factors and these 3 experiments are conducted in different environments.

Table 1: Elevation measurement errors in different mountains

Mountain	Peak Elevation (m)	MAE of DR (m)	MAE of KF (m)	RMSE of KF (m)	MRE of KF (%)
Mt#1	1859	4.63	4.36	5.44	0.3
Mt#1*	1859	4.74	4.46	5.54	0.3
Mt#2	599	3.72	3.74	5.14	1.1
Mt#2*	599	4.69	4.75	6.29	1.3
Mt#3	854	6.07	5.98	10.83	0.9

*: Does not integrate DEM reference

Figure 10 shows DEM elevation, GPS elevation, estimated elevation by our system and air pressure during mountain climbing ($Mt\#1$). Even though we don't know the real elevation profile, **we still can claim our obtained elevation profile is more reliable than GPS and DEM because it clearly looks more natural in mountain climbing**.

Case 3: Inside buildings

In this case, we conducted experiments in 2 different buildings (says $Bld\#1, Bld\#2$ for convenience), which

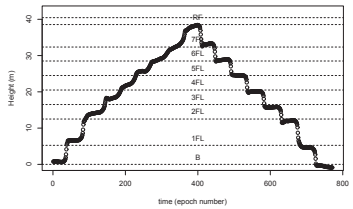


Figure 11: Indoor height measurement

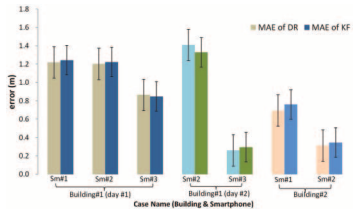


Figure 12: MAE of floor height on different smartphones in different buildings

have 8 floors, 15 floors respectively. Participants took stairs to go up and down in buildings $Bld\#1, Bld\#2$. Inside buildings, there is no good GPS signals, consequently no $DEM\ reference$ is involved. We only evaluate the relative height of every floor by comparing with buildings' floor map. Figure 11 shows the height (relative to the basement floor) measured by our method in building $Bld\#1$. **From which, we find that the estimated height is equal to the floor map when going downstairs (right part of the figure), while it doesn't match the floor map perfectly when going upstairs(left part)**. Although we did not identify the cause of this difference, it is considered that going upstairs involves more irregular movement than going downstairs.

Figure 12 shows the MAE on different smartphones in different buildings ($Bld\#1, Bld\#2$). Here MAE is the error between our obtained height (relative elevation) and the height from floor map on every floor. We found the error is slightly different among smart-phones (different barometer). **The average error is about 0.86 meter**. For building $Bld\#1$, we did experiments on different days ($Bld\#2$ is on another day). We found the average accuracy on day #1 is marginally worse than on day #2. The probable reason is that the weather on day #1 is windy while that on day #2 is sunny.

Related Work

Fusing barometer and GPS to improve the navigation system of aircrafts and UAV (Unmanned Aerial Vehicle) has been studied in various earlier works [9, 4]. Usually airports broadcast reference information, as such barometer based elevation measurement is accurate nearby airport. However, these technologies cannot be directly applied on smartphones in urban areas because reference point may be not within acceptable range and barometer of smartphones performs poorly.

Numerous recent research works [15, 5] have focused on indoor localization where GPS signals are not available, and 3D navigation. Even before the advent of smartphone, some research works [1, 3] had attempted to integrate barometer with mobile platform (e.g. laptop, sensor board) for floor localization. However, they conducted very simple experiments and only worked on relative height, which is less challenging because the error of relative height mainly depends on the quality of barometers. Recent research works [10, 8] have attempted to utilize the embedded barometer sensors in smartphones for measuring height. Li et al.[10] used airport control towers as reference points, and reported the accuracy is reliable only nearby airport, while our method provided reliable result in broader area. This work [16] developed an algorithm to detect floor transition using barometric pressure sensor and accelerometer. Keller et al.[8] did good experiments which revealed that the barometers of smartphones are unstable and less accurate compared to high-performing barometers. It is also reported in [8] that it is necessary to calibrate barometer sensor with respect to temperature, which we have not considered in our method yet.

Conclusion

We presented *iBaro-altimeter*, a system for accurate elevation measurement using barometers on smartphones. To the best of our knowledge, this is the first time to measure elevation (absolute height) with high accuracy using smart-phone. A series of experiments conducted in both indoor and outdoor environments with varying geographic characteristics

using different models of smartphones, reveal that the errors are less than 3 meters in outdoor walking, 6 meters in mountain climbing, 0.9 meter in indoor floor localization. These errors are acceptable for most practical applications.

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